

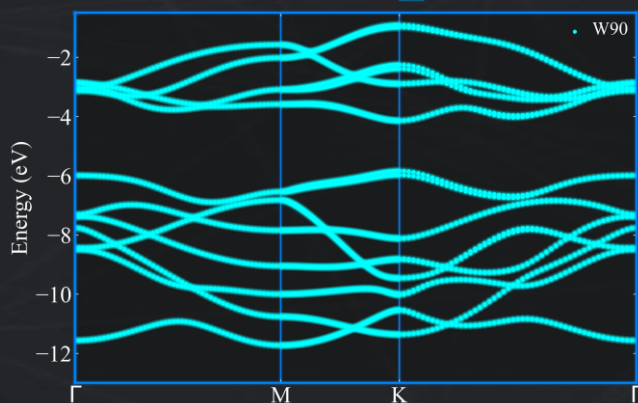
WannierBSE

Zero-Effort Setup: Direct Integration with Wannier90

Connect your Wannier90 results, tune your simulation parameters, and power your discovery with **WBSE.m**

1 Bring your Wannier90 files

wannier90.win
wannier90_*.xsf
wannier90_hr.dat



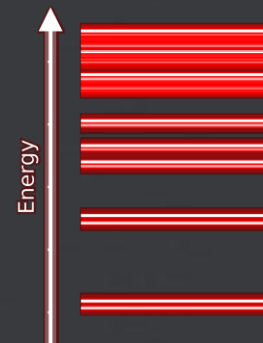
2 Set Simulation Parameters

- control.txt
- exchange_control.txt
- kmesh_control.txt
- WTB_control.txt
- WF_centers.txt
- dielectric_control.txt

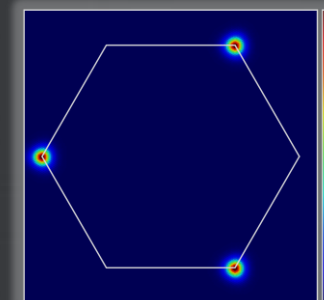
Note: Requires MATLAB
Parallel Computing Toolbox

3 Run **WBSE.m**

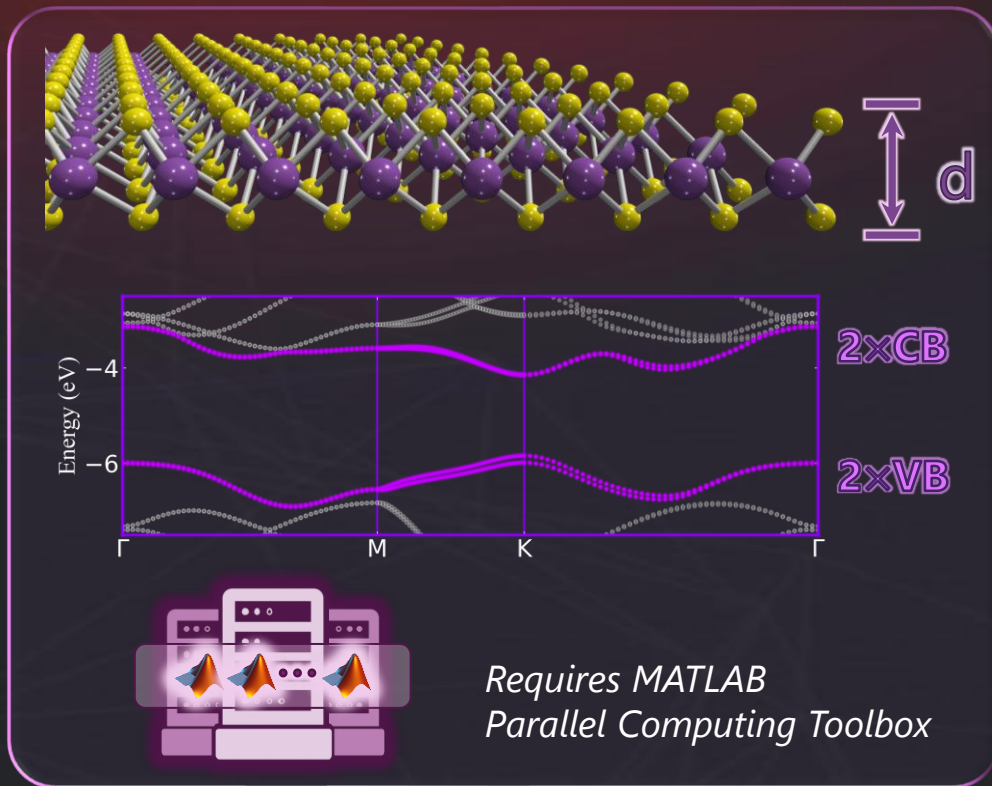
Exciton Energy Spectrum



Exciton Wave Functions



Define the computational settings for the BSE calculation via `control.txt`



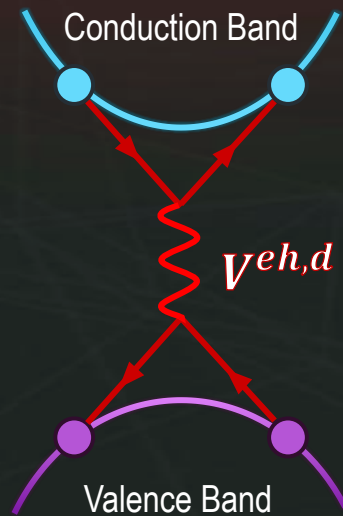
`control.txt`

```
d = 6.26           % 2D material thickness in angstroms
%--- BSE calculation options ---
NVAL = 100         % # of exciton eigenvalues to find
Nc = 2             % Conduction bands in BSE
Nv = 2             % Valence bands in BSE
%--- Parallel calculation parameter ---
Npar_TB   = 12     % # of cores for TB band generation
Npar_rho  = 12     % # of cores for exchange density
Npar_Xi   = 12     % # of cores for SR Xi integrals
Npar_HBSE = 12     % # of cores for BSE construction
```

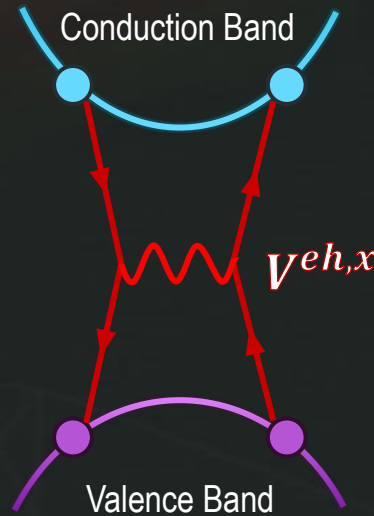
Input monolayer thickness (d), select your active band window (Nc, Nv), and set your target eigenvalues. Accelerate each calculation stage through parallel processing

Enable short-range exchange through `exchange_control.txt`

Direct



Exchange



`exchange_control.txt`

```
Exchange_Interaction = true % Enable SR exchange (G~=0)
Use_Precomputed_Chi = true % Load precomputed chi
cleanup_xsf          = false % Keep raw XSF files
```

% --- Exchange Parameters ---

```
M_max      = 1 % Maximum neighbor shell
G_max      = 4.2 % Exchange G-vector cutoff
tol_probability = 0.98 % Wannier-box probability cutoff
M_max_tolerance = 0.01 % Neighbor-shell distance tolerance
nn_distance_tolerance = 0.15 % NN classification tolerance
```



Enable exchange interaction

Set `Exchange_Interaction = true`



Optionally reuse precomputed chi

Set `Use_Precomputed_Chi = true`

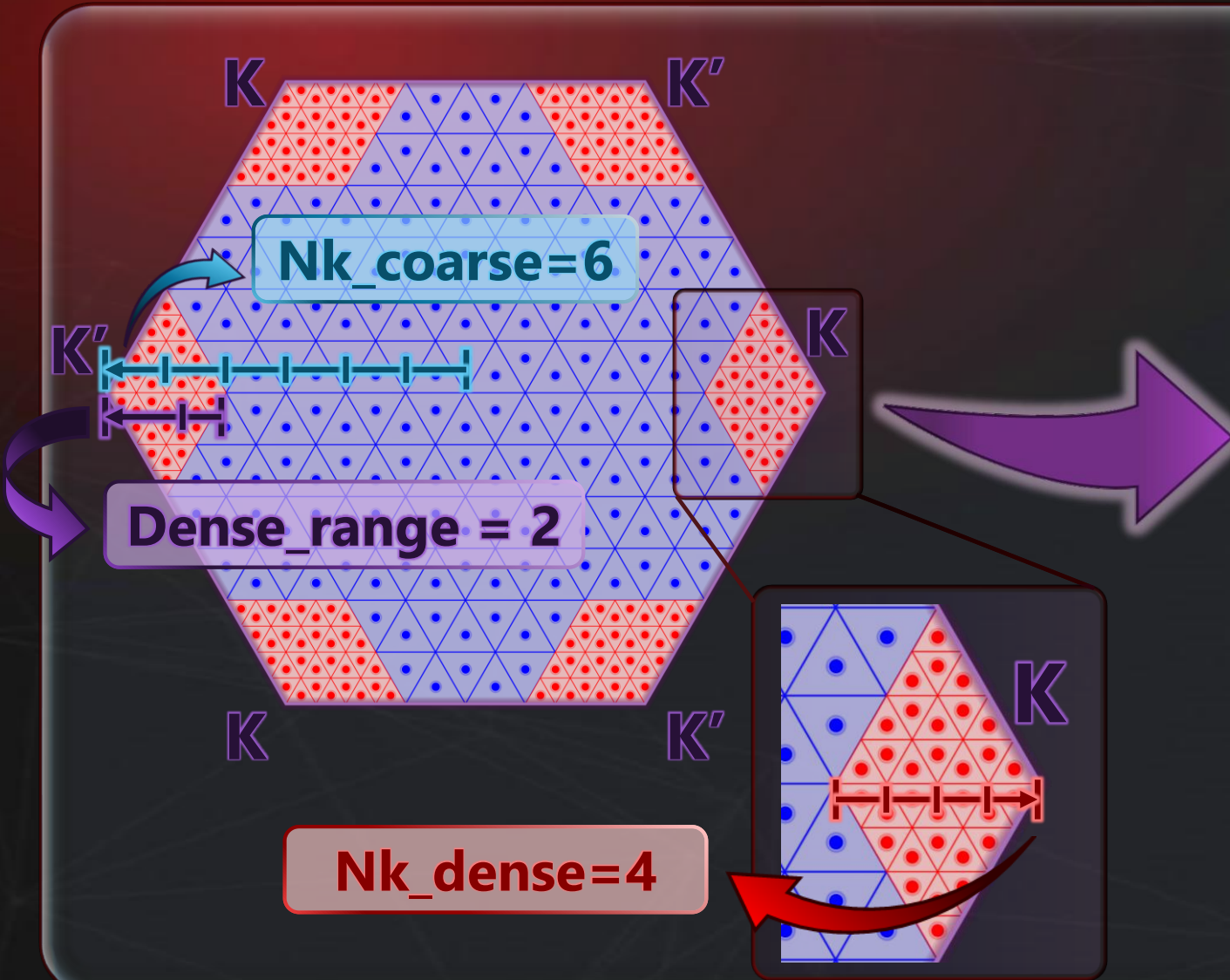


Keep XSF files

Set `cleanup_xsf = false`

WannierBSE

Set your non-uniform k-mesh



`kmesh_control.txt`

% --- Control parameters for k-mesh generation ---

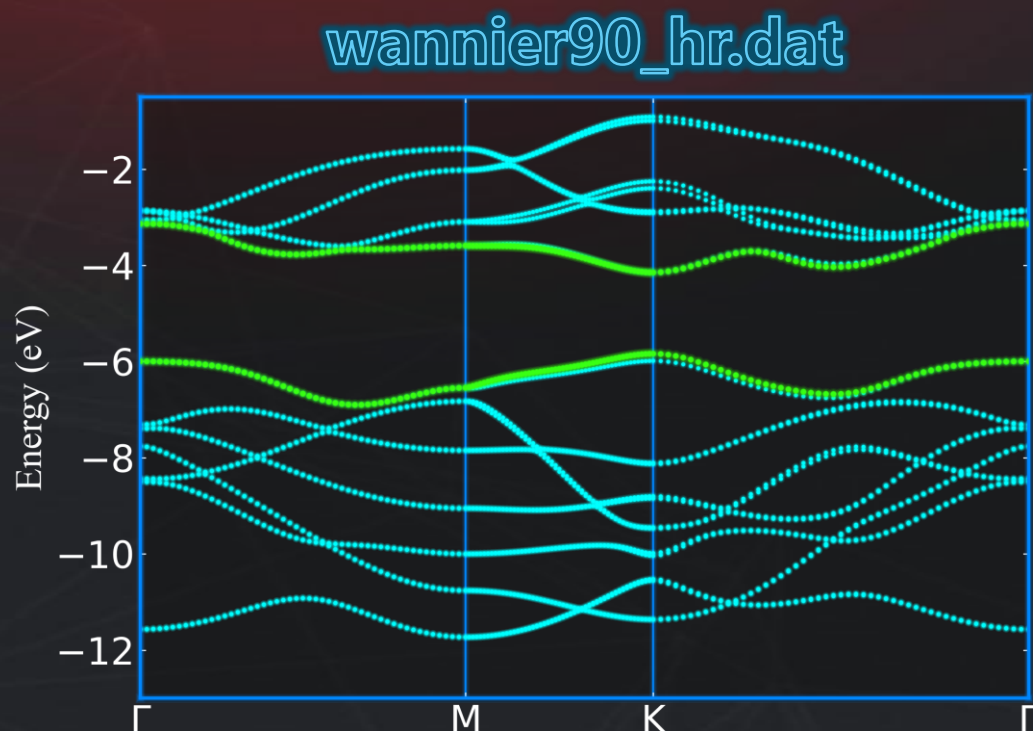
`Nk_coarse = 6` % K-points per direction in coarse region

`Nk_dense = 4` % K-points per direction in dense region

`Dense_range = 2` % Dense-region size in coarse-grid units



Configure your **WTB_control.txt** file directly from your **Wannier90** results



WTB_control.txt

%--- Control parameters for Wannier Tight-Binding

VBM = 14 % Top valence band index
CBM = 15 % Lowest conduction band index

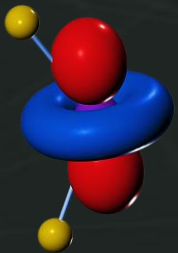
Select the top valence
and lowest conduction bands
from your wannier90_hr.dat output.

Assign each Wannier function to its atomic center through **WF_centers.txt**



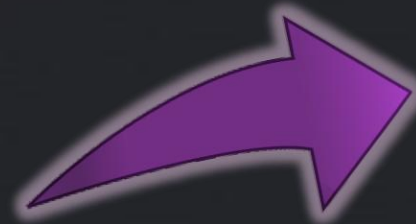
wannier90.wout

Wannier centers and
nearest atomic sites



Plot Wannier
Functions

Visual localization verification



WF_centers.txt

% i-th Wannier Function center
at τ_I atom

i	I	Spinor
1	2	up
2	2	down
3	2	up
4	2	down
5	2	up
...		

1

WF index

2

Associated atomic index

3

Spinor Label: up or down



dielectric_control.txt

```
% --- Material Parameters for the TMD-ML
alpha = 1.55    % Fitting parameter
Epl  = 22.5     % Plasmon peak energy (eV)
ep_p  = 13.24   % In-plane dielectric constant
```

```
% --- Number of Layers
t_layers = 2    % layers ABOVE the TMD-ML
s_layers = 1    % layers BELOW the TMD-ML
```

```
% --- Dielectric Constants
t_dielectric = 1, 5.89
s_dielectric = 3.9
```

```
% --- Boundaries (Å)
layer_boundaries = 103.13, 3.13, -3.13
```

